

5.

Izračunaj eksponente

$$a) \quad i\varphi\sigma_z = \begin{bmatrix} i\varphi & 0 \\ 0 & -i\varphi \end{bmatrix}$$

$$e^{i\varphi\sigma_z} = \begin{bmatrix} e^{i\varphi} & 0 \\ 0 & e^{-i\varphi} \end{bmatrix}$$

$$b) \quad i\varphi\sigma_x = \begin{bmatrix} 0 & i\varphi \\ i\varphi & 0 \end{bmatrix}$$

$$P_{\sigma_x}(\lambda) = \lambda^2 + \varphi^2 = (\lambda + i\varphi)(\lambda - i\varphi)$$

$$D = i\varphi \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$P = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$e^{i\varphi\sigma_x} = P e^D P^{-1} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} e^{i\varphi} & 0 \\ 0 & e^{-i\varphi} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} e^{i\varphi} & \frac{\sqrt{2}}{2} e^{i\varphi} \\ -\frac{\sqrt{2}}{2} e^{-i\varphi} & \frac{\sqrt{2}}{2} e^{-i\varphi} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(e^{i\varphi} + e^{-i\varphi}) & \frac{1}{2}(e^{i\varphi} - e^{-i\varphi}) \\ \frac{1}{2}(e^{i\varphi} - e^{-i\varphi}) & \frac{1}{2}(e^{i\varphi} + e^{-i\varphi}) \end{bmatrix} =$$

$$= \begin{bmatrix} \cos\varphi & i\sin\varphi \\ i\sin\varphi & \cos\varphi \end{bmatrix}$$

⑥ z uporabo Cayley-Hamiltonovega izreka izračunaj eksponent matrike

$$W = \begin{bmatrix} 0 & -\phi \\ \phi & 0 \end{bmatrix}$$

$$P_W(\lambda) = \lambda^2 + \phi^2$$

$$W^2 = -\phi^2 I$$

(Greš na vrste, se mi ni dal prepisovat)

$$e^W = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

⑦ $T: \mathbb{C}^n \rightarrow \mathbb{C}^n$ endomorf., ima same razl. last. vr.

Pokaži da je vsak end., ki z njim komutira, diagonalizabilen.

$-T$ je diag.

$$\rightarrow [T]_B^B = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \quad i \neq j \Rightarrow \lambda_i \neq \lambda_j$$

$$[X]_B^B = \begin{bmatrix} x_{11} & \dots & x_{1n} \\ \vdots & & \vdots \\ x_{n1} & \dots & x_{nn} \end{bmatrix}$$

$$[TX]_B^B = \begin{bmatrix} \lambda_1 x_{11} & \dots & \lambda_1 x_{1n} \\ \lambda_2 x_{21} & \dots & \lambda_2 x_{2n} \\ \vdots & & \vdots \\ \lambda_n x_{n1} & \dots & \lambda_n x_{nn} \end{bmatrix}$$

$$[XT]_B^B = \begin{bmatrix} \lambda_1 x_{11} & \dots & \lambda_n x_{1n} \\ \vdots & & \vdots \\ \lambda_1 x_{n1} & \dots & \lambda_n x_{nn} \end{bmatrix}$$

$$TX = XT \xrightarrow{i \neq j} \lambda_i x_{ij} = \lambda_j x_{ij} \Rightarrow x_{ij} = 0$$

$$X = \begin{bmatrix} x_{11} & & \\ & x_{22} & \\ & & \ddots \\ & & & x_{nn} \end{bmatrix} \Rightarrow \underline{X \text{ je diag. (v isti bazi kot } T)}$$

VELJA TUDI ZA BLOČNO DIAG. T

$$T = \begin{bmatrix} \lambda_1 I_{k_1} & & \\ & \ddots & \\ & & \lambda_n I_{k_n} \end{bmatrix}$$

⑧ Naj bosta $A, B: \mathbb{C}^3$ diag. end.

Pokaži, da $[AB] = 0 \Leftrightarrow \exists$ baza, kjer pripadata A in B diag. matriki

($\Leftarrow$)

$$[A]_B^B = \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{bmatrix}$$

$$AB = \begin{bmatrix} a_1 b_1 & & \\ & a_2 b_2 & \\ & & a_3 b_3 \end{bmatrix}$$

$$[B]_B^B = \begin{bmatrix} b_1 & & \\ & b_2 & \\ & & b_3 \end{bmatrix}$$

$$BA = \begin{bmatrix} b_1 a_1 & & \\ & b_2 a_2 & \\ & & b_3 a_3 \end{bmatrix}$$

$$AB - BA = 0 \checkmark$$

($\Rightarrow$)

$$1.) [A]_B^B = \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{bmatrix}$$

Prejšnja naloga

$$2.) [A]_B^B = \begin{bmatrix} a_1 & a_1 & \\ & a_1 & \\ & & a_1 \end{bmatrix} = a_1 I$$

Vsaka mat. B komutira z $a_1 I$

$$3.) [A]_B^B = \begin{bmatrix} a_1 & a_1 & \\ & a_1 & \\ & & a_2 \end{bmatrix}$$

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = a_1 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow \begin{aligned} x a_1 &= a_1 x \\ y a_1 &= a_1 y \\ z a_2 &= a_1 z \rightarrow z = 0 \end{aligned}$$

last. vekt. pri a_1 : $\begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \rightarrow v$ (ved.)

last. vekt. pri a_2 : $\begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} \rightarrow u$ (narp.)

$$B = \begin{bmatrix} b_{11} & b_{12} & 0 \\ b_{21} & b_{22} & 0 \\ 0 & 0 & b_{33} \end{bmatrix}$$

$AB = BA$

$$B \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ 0 \end{bmatrix}$$

$$B \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ z' \end{bmatrix} \quad z' = b_{33} z$$

so last. vekt.

$$\left. \begin{aligned} x' &= x b_{11} + y b_{12} \\ y' &= x b_{21} + y b_{22} \end{aligned} \right\} \text{ niso nujno last. vekt.}$$

u je last. vekt. za B

$u_v + u_n$

$$Bu = Bu_v + Bu_n = \lambda u_v + \lambda u_n$$

$$B u_v = \lambda u_v \quad B u_u = \lambda u_u$$

u_v, u_u sta lastna vektorja B

$$[B]_{B'}^{B'} = \text{diag}$$

$$B' = \{u_1, u_2, u_3\}$$

$$u_3 \mapsto \begin{pmatrix} 0 \\ 0 \\ (u_3)_3 \end{pmatrix} \quad \leftarrow \text{še vedno last. vekt.}$$

$$u_1 \mapsto \begin{pmatrix} (u_1)_1 \\ (u_1)_2 \\ 0 \end{pmatrix}, \quad u_2 \mapsto \begin{pmatrix} (u_2)_1 \\ (u_2)_2 \\ 0 \end{pmatrix}$$

v B' je tudi A diagonalna